

Explain why market failure occurs in the case of public goods, where the market is without government intervention

Market failure occurs when a free market does not allocate resources efficiently. Due to the unique characteristics of public goods, where public goods are non-excludible and non-rivalrous, producers would not be incentivised to produce public goods assuming the producers in the market have profit maximisation as their primary goal.

For example, a streetlight that can be found on the side of a public road is a public good. When an individual uses the streetlight, the amount of light that is emitted by the streetlight remains the same, and its quality is similarly unaffected by the consumer; the streetlight is non-rivalrous, since consumption of said good does not diminish the quality or quantity of the streetlight for the next consumer. Hence, the marginal cost — the additional costs of producing one more unit of output — would be 0. In order to achieve allocative efficiency, producers would have to sell streetlights to consumers at the amount it cost to make; so it can be concluded that producers would not be able to charge consumers a price for street lighting and there will be zero allocation of resources in the production of such a public good, leading to market failure.

Additionally, it is not economically feasible or practical to exclude non-payers from consuming light emitted by a streetlight; it is non-excludible. Once a streetlight has been produced, those who have not paid for it are still able to enjoy its lighting. Therefore, there is no incentive for consumers to reveal their demand, leading to the free-rider problem. And so, there would be no price signal for firms to produce such a good as it is unprofitable, meaning that there would be no resources allocated to its production. As a result, market failure occurs.

In summary, the example of a streetlight in this response illustrates the non-rivalrous and non-excludible nature of all public goods, where firms in the private sector lack the incentive to allocate resources for, or to produce, such goods, leading to market failure. Hence, governments must intervene to provide public goods.